School of Computing

21CSE306P · APPLIED GENERATIVE AI

Industry Supported Professional Elective · B.Tech CSE, 5th Semester · Dept. of Networking and Communications

SRM INSTITUTE OF SCIENCE & TECHNOLOGY

4 QUALITY EDUCATION

AI-Powered Video Summarization for Smart Learning

Ask a lecture — or an entire course playlist — and get answers grounded in what was actually said

PROBLEM STATEMENT & OBJECTIVE

- A one-hour lecture cannot be searched — re-watching is the only option
- A course is many videos; a topic is taught across several of them
- Generic AI summaries invent facts the lecture never stated

OBJECTIVES

- Turn a video, or a whole playlist, into a searchable study guide
- Every claim traceable to a real moment in a real video
- Answer questions using only what was actually said

SOLUTION

- **Retrieval-Augmented Generation** — only retrieved transcript reaches the model
- **Course mode** — one shared index across a playlist, answers cite video + timestamp
- **Timestamp validation** — any moment not in the retrieved text is discarded
- **Grounded refusal** — returns "not covered in this video" instead of guessing

METHODOLOGY — PIPELINE

1. **Ingestion** — YouTube Transcript API
   timed transcript
2. **Chunking** — 1000 chars · 200 overlap
   overlapping chunks
3. **Embedding** — all-MiniLM-L6-v2 → ChromaDB
   384-d vectors
4. **Retrieval** — top-k 5, re-ranked
   grounding context
5. **Generation** — open-mistral-nemo
   JSON draft
6. **Output** — Key points · timings · mind map

failures & retries | *prompts & replies* | *finished summary*

7 Error handling & validation

8 Red teaming — injection, safety, bias

9 Evaluation & human review (offline)

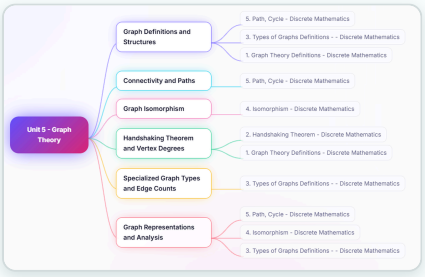
Stages 1–6 run in fixed order; layers 7–9 apply across all of them. Stage 9 is reviewed offline — scores never change a setting on their own.

GENAI MODEL & TECHNIQUES

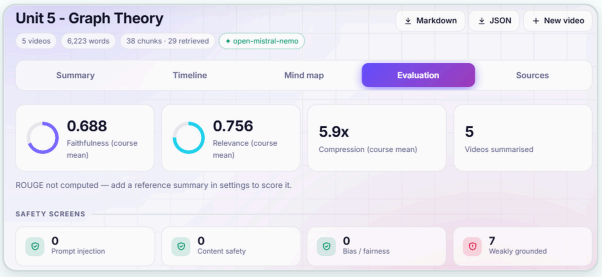
Generation by **open-mistral-nemo** (Mistral) · embeddings by **all-MiniLM-L6-v2** · vector store **ChromaDB**

- Retrieval-Augmented Generation
- Role prompting
- Grounding constraint
- Structured output (JSON schema)
- Few-shot exemplar

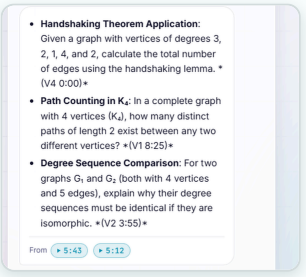
OUTPUT & RESULTS



Mind map across the whole playlist



Course evaluation — 5 videos, 38 chunks indexed together



Answers cite the video and the moment (V4 0:00, V1 8:25)

0.688	0.756	5.9x	5
FAITHFULNESS	RELEVANCE	COMPRESSION	VIDEOS SUMMARISED

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